

Information on presentation

Each subgroup will showcase their individual wind turbine model. The prof.s will then select one model from each group for production based on the evaluation of the presented model characteristics.

When:

12-13/10. Registration on the google sheet.

Duration:

- 15min presentation of both subgroups belonging to the same group.
- 12min discussion.
- 3min to move to next group.

Two groups listen to each other. The purpose is to learn also from what the others did.

The following points must be included in the presentation and will be considered for the selection of the model:

Blades:

- CAD model of the whole rotor, showing clearly the orientation of the two blades.
- Show that the rotor blades adhere to the following constraints:
 - o Chord length is $40mm < L_c < 100mm$ along the whole blade
 - o Maximum airfoil thickness at each element is at least 5mm along the whole blade
 - o Absolute min thickness of 0.6mm all over the blade
- Efficiency (C_p) at the design point
- Methodology

Generator (both groups must have the same wire and generator type, in terms of nr. poles and phases):

- Active length of coil side
- Number of turns per coil (N)
- Number of coils in series (Ns)
- Number of phases (Nphase)
- Number of poles (P)
- Figure with the coil layout
- Copper wire kind and diameter (Dcu)
- Total weight of coils
- Efficiency at design point
- Methodology

General parameters:

- Design RPM and TSR
- Torque difference between rotor blades and generator
- Total efficiency

Choice of the model:

Prof.s and the scientific assistant will gather after all the presentations and decide which of the two models from each group will be produced, following those criteria:

		POINTS (1-10)
GENERATOR	Best numerical efficiency	
	Lightest	
	Creative Design/Methodology	
BLADES	Best numerical efficiency	
	Creative Design/Methodology	
SYSTEM	Total best numerical efficiency	
SUM		

GROUP evaluation of the presentation:

The presentation counts for 25% of the overall grade, assessed collectively for the entire group. Note that the grade won't be given until the report will be graded as well.

Grading criteria:

- **Presentation quality:** are the slides informative and logically organized? Is the presentation clear and engaging?
- **Language proficiency:** is the language used precise and accurate? Is the communication effective?
- **Subject matter understanding:** are the questions addressed appropriately? Does the discussion show knowledge of the subject matter?
- **Collaboration:** is there clear active participation and cooperation from all group members? Not necessarily all the group members must present, since there is not much time for that, but each group member should either present and/or be active in the discussion, and show knowledge and engagement in the project.

Deadline for delivering presentation and CAD files

The presentation of each group must be uploaded on Canvas by Saturday 10th October.

Name the files:

- "NN_Presentation", where NN=group. One presentation for each group means that contains both the subgroups. Each subgroup presents, but the two presentations are merged in one file, either .pptx or .pdf.

The CAD model of each subgroup must be sent to Elias Öhman <elias.ohman@ntnu.no> by Thursday 08th October.

Name BOTH email subject and file:

- "NNm_CADmodel", where NN=group number and m=subgroup (either A or B). The cad file must be .STEP.